Number System

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SWE 4101
Introduction to Software Engineering

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Outline

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- Non-positional number system.
- Positional number system.
- Decimal number system.
- Binary number system.
- Octal number system.
- Hexadecimal number system.

Number System

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- Two types of number system
  - Non-positional number system.
  - Positional number system.

Decimal Number System

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- Positional number system.
- Digits – 0,1,2,3,4,5,6,7,8,9.
- Example:

$$4321_{10} = 1 * 10^0 + 2 * 10^1 + 3 * 10^2 + 4 * 10^3$$
$$= 1 + 20 + 300 + 4000$$

Binary Number System

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- Positional number system.
- Digits – 0,1.
- Example:

$$1010_2 = 0 * 2^0 + 1 * 2^1 + 0 * 2^2 + 1 * 2^3$$
$$= 0 + 2 + 0 + 8$$

Octal Number System

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- Positional number system.
- Digits – 0,1,2,3,4,5,6,7.
- Example:

$$7654_8 = 4 * 8^0 + 5 * 8^1 + 6 * 8^2 + 7 * 8^3$$
$$= 4 + 40 + 384 + 3584$$
$$= 4012$$

Converting Decimal Value To Binary

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to Base	Number (Quotient)	Remainder
2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1



$(25)_{10} = (11001)_2$

Fractional Number

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- The positional value of the digit of a number after fractional point is

$$X = base^{-position}$$

[Position is counted from the fraction point]

						Binary Point ↓			
Position	4	3	2	1	0	.	-1	-2	-3	-4
Position Value	2^4	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	2^{-4}
Quantity Represented	16	8	4	2	1		$^1/_2$	$^1/_4$	$^1/_8$	$^1/_{16}$

Fractional Number

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- Decimal from octal number

$$\begin{aligned} 127.54_8 &= 1 \times 8^2 + 2 \times 8^1 + 7 \times 8^0 + 5 \times 8^{-1} + 4 \times 8^{-2} \\ &= 64 + 16 + 7 + {}^5/_8 + {}^4/_{64} \\ &= 87 + 0.625 + 0.0625 \\ &= 87.6875_{10} \end{aligned}$$

Shortcut method (Binary to Octal)

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$1101010_2 = ?_8$

Step 1: Divide the binary digits into groups of 3 starting from right

001 101 010

Step 2: Convert each group into one octal digit

$001_2 = 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1$
$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5$
$010_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$

Hence, $1101010_2 = 152_8$

Shortcut method (Octal to Binary)

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$562_8 = ?_2$

Step 1: Convert each octal digit to 3 binary digits

$5_8 = 101_2$, $6_8 = 110_2$, $2_8 = 010_2$

Step 2: Combine the binary groups

$562_8 =$ 101 110 010
5 6 2

Hence, $562_8 = 101110010_2$

Shortcut method (Binary to Hexadecimal)

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$111101_2 = ?_{16}$

Step 1: Divide the binary digits into groups of four starting from the right

0011 1101

Step 2: Convert each group into a hexadecimal digit

$0011_2 = 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 3_{10} = 3_{16}$
$1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 3_{10} = D_{16}$

Hence, $111101_2 = 3D_{16}$